Chapter 6: Screening Designs

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1 Introduction

This chapter covers screening designs including sensitivity analysis and Morris screening designs.

Citation: V. R. Joseph (2026). *Experimental Design for Data Science and Engineering*. CRC Press.

```
library(sensitivity)
library(SFDesign)
library(rkriging)
```

2 Section 6.1: Sensitivity Analysis

2.1 Borehole Function

```
# Borehole function
f <- function(x) {
  lower <- c(0.05, 100, 63070, 990, 63.1, 700, 1120, 9855)
  upper <- c(0.15, 50000, 115600, 1110, 116, 820, 1680, 12045)
  x <- lower + x * (upper - lower)
  val <- 2 * pi * x[3] * (x[4] - x[6]) /
    (log(x[2] / x[1]) * (1 + 2 * x[7] * x[3] / (log(x[2] / x[1]) * x[1]^2 * x[8]) + x[3] / x[5]))
  return(val)
}
```

2.2 6.1.1 Variance-based Sensitivity Measures

```r
set.seed(1)
p <- 8
m <- 10000

A <- matrix(runif(m * p), nrow = m)
B <- matrix(runif(m * p), nrow = m)
borehole <- function(X) apply(X, 1, f)

a.sen <- soboljansen(model = borehole, X1 = data.frame(A), X2 = data.frame(B))
sf <- pmax(unlist(a.sen$S), 0)
st <- pmax(unlist(a.sen$T), 0)

x <- matrix(c(sf, st - sf), nrow = 2, byrow = TRUE)
colnames(x) <- paste("x", sep = "", 1:p)
rownames(x) <- c("First Order", "Total - First Order")

par(mar = c(5, 5, 4, 2))
par(mfrow = c(1, 2))

barplot(x, col = c("lightblue", "pink"), legend.text = TRUE,
        cex.names = 1.5, args.legend = list("cex" = 1.5, "bty" = "n"),
        ylab = "Total Sobol' Index", cex.lab = 1.5, main = "Sobol' Indices")

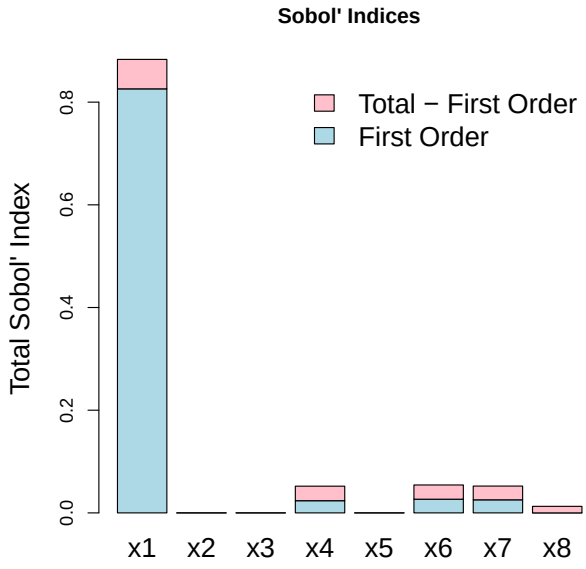
# Main effects using GP emulator
D <- maxpro.optim(maxproLHD(10 * p, p)$design)$design
y <- apply(D, 1, f)
a.fit <- Fit.Kriging(D, y, kernel.parameters = list(type = "Gaussian"))
fhat <- function(x, fit) Predict.Kriging(fit, matrix(x, ncol = p))$mean

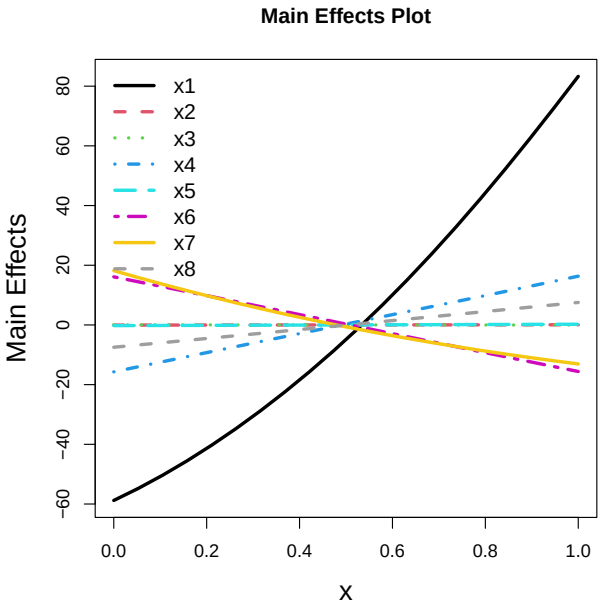
N <- 10 * p
D.eval <- uniformLHD(N, p)$design
f.eval <- apply(D.eval, 1, fhat, a.fit)
f0 <- mean(f.eval)
x.ev <- seq(0, 1, length = 20)

main.effect <- function(ind) {
  D0 <- D.eval
  val <- numeric(20)
  for (i in 1:20) {
    D0[, ind] <- rep(x.ev[i], N)
    val[i] <- mean(apply(D0, 1, fhat, a.fit))
  }
  return(val - f0)
}

M <- matrix(0, nrow = 20, ncol = p)
for (j in 1:p) M[, j] <- main.effect(j)

matplot(x.ev, M, type = "l", lty = 1:p, col = 1:p, lwd = 3,
        xlab = "x", ylab = "Main Effects", cex.lab = 1.5,
        main = "Main Effects Plot")
legend("topleft", paste0("x", 1:p), bty = "n", lty = 1:p, col = 1:p, lwd = 3, cex = 1.2)
```





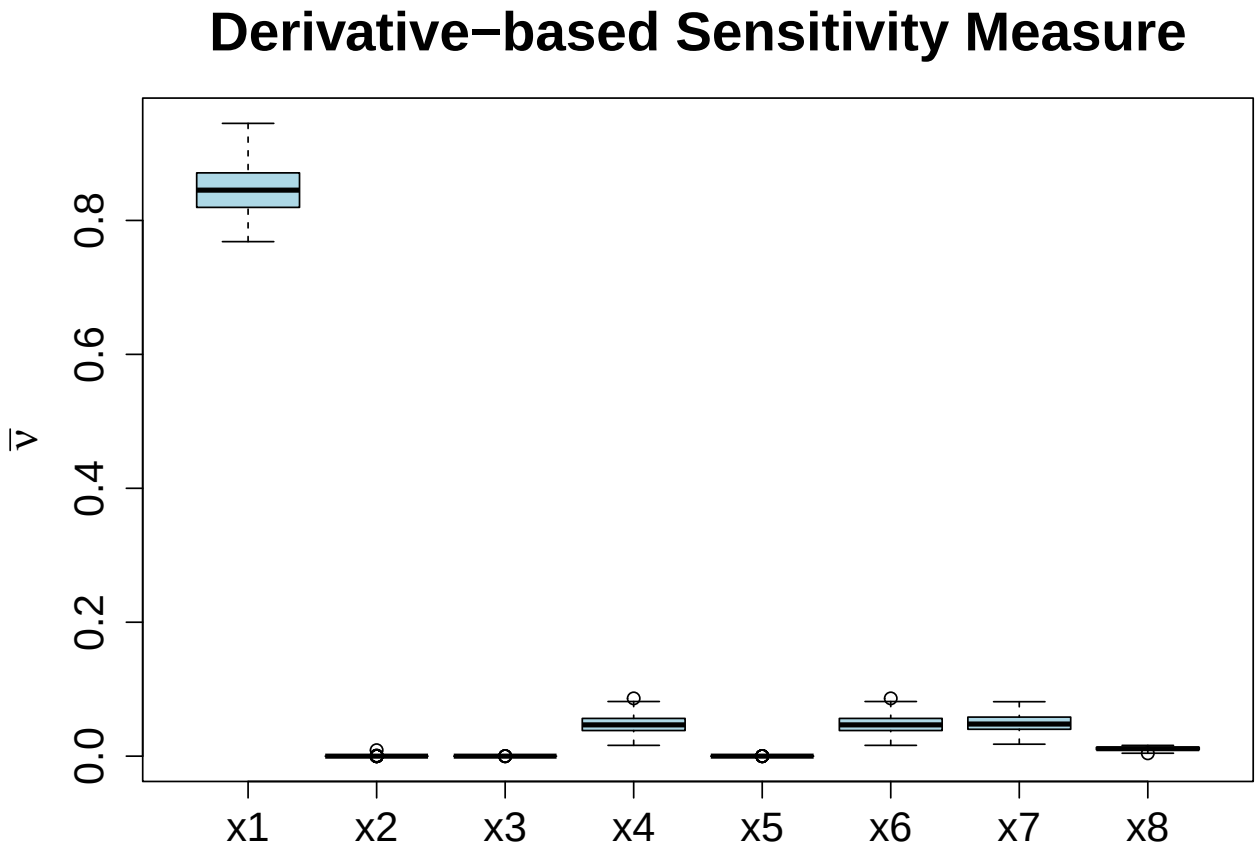

```
par(mfrow = c(1, 1))
```

2.3 6.1.2 Derivative-based Sensitivity Measures

```
set.seed(1)
p <- 8
m <- 4
nu <- matrix(0, nrow = 100, ncol = p)

for (i in 1:100) {
  X <- matrix(runif(m * p), ncol = p)
  ad <- delsa(model = borehole, X0 = X, varprior = rep(1, p))
  nu[i, ] <- colMeans(ad$deriv^2)
  nu[i, ] <- nu[i, ] / sum(nu[i, ])
}

lab <- paste("x", sep = "", 1:p)
colnames(nu) <- lab
boxplot(nu, main = "Derivative-based Sensitivity Measure",
        cex.main = 2, cex.axis = 1.5, col = "lightblue",
        ylab = expression(bar(nu)), cex.lab = 1.5)
```



3 Section 6.2: Morris Screening Design

3.1 Morris Design Illustration

```
set.seed(10)
par(mar = c(5, 5, 4, 2))
par(mfrow = c(1, 2))

x <- ((1:4) - 1) / 3
A <- cbind(x, sample(x))
delta <- 1/3

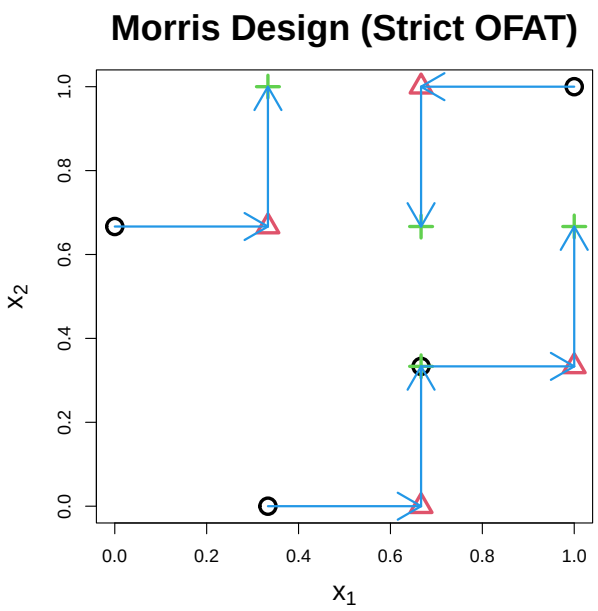
# Strict OFAT
C1 <- C2 <- A
for (i in 1:4) {
  C1[i, 1] <- A[i, 1] + delta
  if (C1[i, 1] < 0 || C1[i, 1] > 1) C1[i, 1] <- A[i, 1] - delta
  C2[i, 1] <- C1[i, 1]
  C2[i, 2] <- C1[i, 2] + delta
  if (C2[i, 2] < 0 || C2[i, 2] > 1) C2[i, 2] <- C1[i, 2] - delta
}

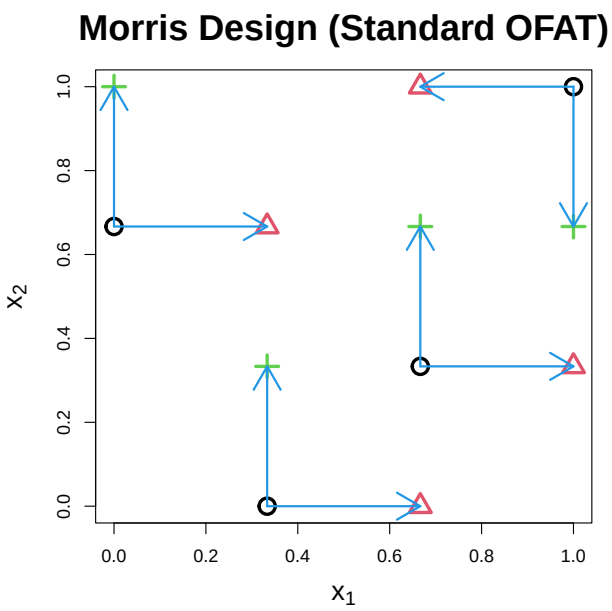
plot(A, xlab = expression(x[1]), ylab = expression(x[2]),
     xlim = c(0, 1), ylim = c(0, 1), lwd = 3, cex = 2,
     main = "Morris Design (Strict OFAT)", cex.main = 2, cex.lab = 1.5)
points(C1, col = 2, pch = 2, lwd = 3, cex = 2)
```

```
points(C2, col = 3, pch = 3, lwd = 3, cex = 2)
for (i in 1:4) {
  arrows(A[i, 1], A[i, 2], C1[i, 1], C1[i, 2], col = 4, lty = 1, lwd = 2)
  arrows(C1[i, 1], C1[i, 2], C2[i, 1], C2[i, 2], col = 4, lty = 1, lwd = 2)
}

# Standard OFAT
C1 <- C2 <- A
for (i in 1:4) {
  C1[i, 1] <- A[i, 1] + delta
  if (C1[i, 1] < 0 || C1[i, 1] > 1) C1[i, 1] <- A[i, 1] - delta
  C2[i, 2] <- C1[i, 2] + delta
  if (C2[i, 2] < 0 || C2[i, 2] > 1) C2[i, 2] <- C1[i, 2] - delta
}

plot(A, xlab = expression(x[1]), ylab = expression(x[2]),
     xlim = c(0, 1), ylim = c(0, 1), lwd = 3, cex = 2,
     main = "Morris Design (Standard OFAT)", cex.main = 2, cex.lab = 1.5)
points(C1, col = 2, pch = 2, lwd = 3, cex = 2)
points(C2, col = 3, pch = 3, lwd = 3, cex = 2)
for (i in 1:4) {
  arrows(A[i, 1], A[i, 2], C1[i, 1], C1[i, 2], col = 4, lty = 1, lwd = 2)
  arrows(A[i, 1], A[i, 2], C2[i, 1], C2[i, 2], col = 4, lty = 1, lwd = 2)
}
```






```
par(mfrow = c(1, 1))
```

3.2 Morris Screening Results

```
set.seed(1)
p <- 8

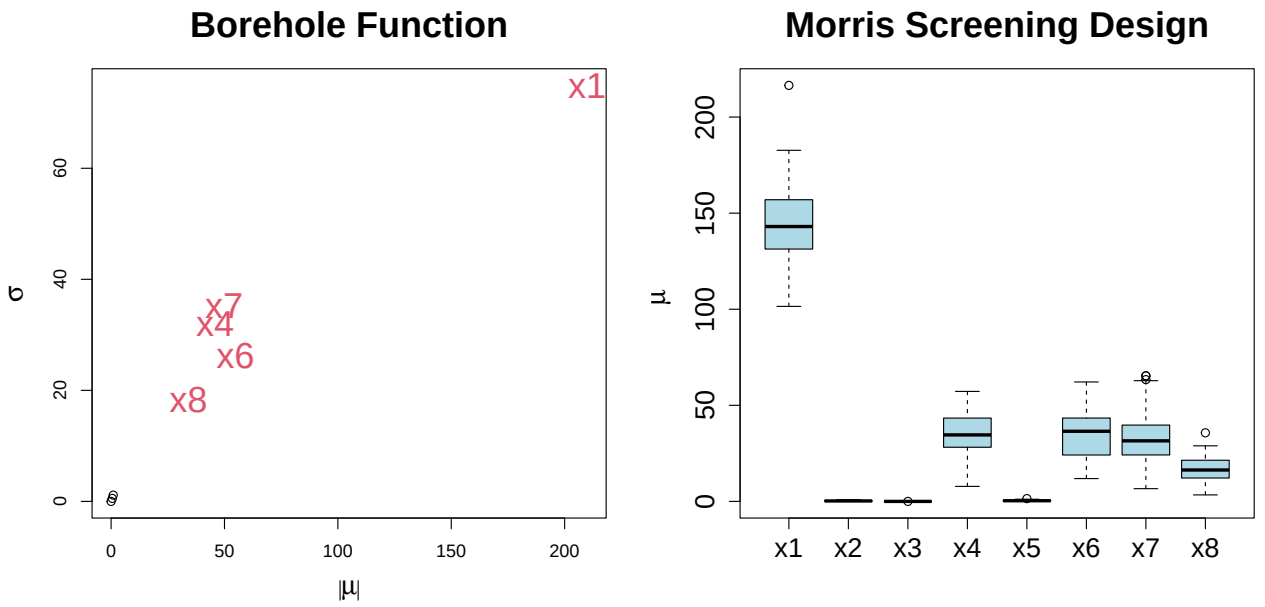
a <- morris(model = borehole, factors = 8, r = 4,
            design = list(type = "oat", levels = 4, grid.jump = 2))
```

```r
mustar <- apply(a$ee, 2, function(x) mean(abs(x)))
sigma <- apply(a$ee, 2, sd)
mu <- abs(apply(a$ee, 2, function(x) mean(x)))

par(mfrow = c(1, 2))

# mu vs sigma plot
sel <- (mu > 10)
lab <- paste("x", sep = "", 1:p)
plot(mu, sigma, type = "n", xlab = expression(abs(mu)), ylab = expression(sigma),
     cex.lab = 1.5, main = "Borehole Function", cex.main = 2, cex = 2)
text(mu[sel], sigma[sel], lab[sel], cex = 2, col = 2)
points(mu[!sel], sigma[!sel])

# Multiple runs boxplot
mustar <- matrix(0, nrow = 100, ncol = p)
for (i in 1:100) {
  am <- morris(model = borehole, factors = p, r = 4,
               design = list(type = "oat", levels = 4, grid.jump = 2))
  mustar[i, ] <- apply(am$ee, 2, function(x) mean(abs(x)))
}
colnames(mustar) <- lab
boxplot(mustar, main = "Morris Screening Design", cex.main = 2,
        cex.axis = 1.5, col = "lightblue", ylab = expression(mu^"*"), cex.lab = 1.5)
```




```r
par(mfrow = c(1, 1))
```

4 Section 6.3: MOFAT Designs

4.1 Sobol Design Illustration

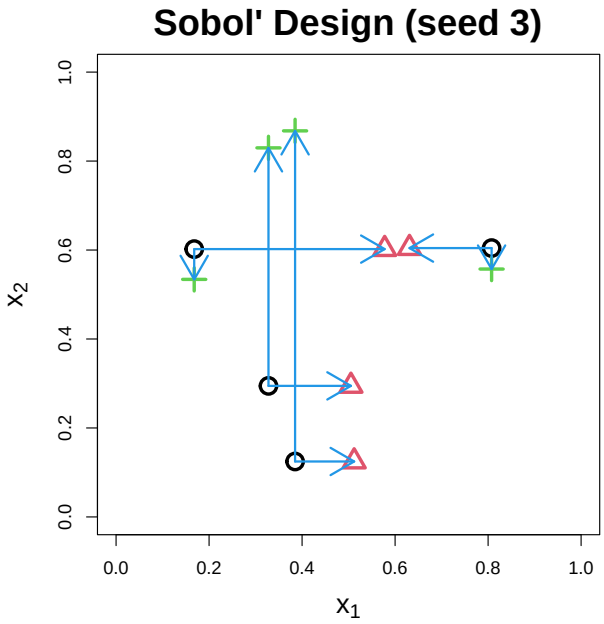
```r
par(mfrow = c(1, 2))
```

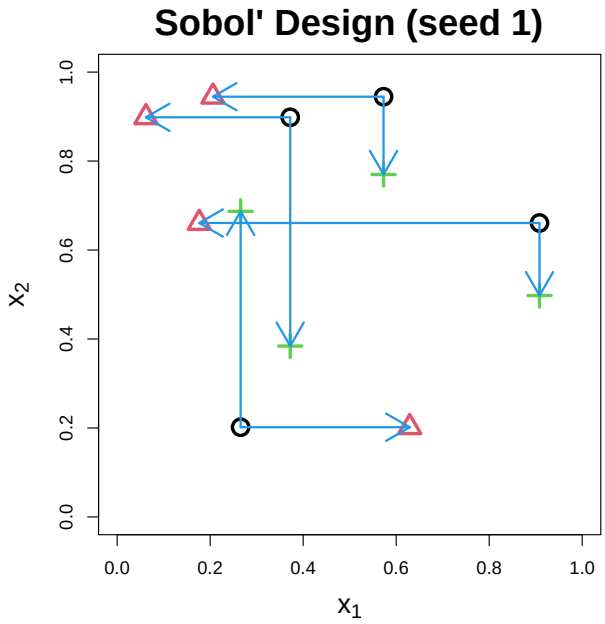
```r
set.seed(3)
par(mar = c(5, 5, 3, 2))
A <- matrix(runif(4 * 2), nrow = 4, ncol = 2)
B <- matrix(runif(4 * 2), nrow = 4, ncol = 2)
C1 <- cbind(B[, 1], A[, 2])
C2 <- cbind(A[, 1], B[, 2])
D <- rbind(A, C1, C2)

plot(A, xlab = expression(x[1]), ylab = expression(x[2]),
     xlim = c(0, 1), ylim = c(0, 1), lwd = 3, cex = 2,
     main = "Sobol' Design (seed 3)", cex.main = 2, cex.lab = 1.5)
points(C1, col = 2, pch = 2, lwd = 3, cex = 2)
points(C2, col = 3, pch = 3, lwd = 3, cex = 2)
for (i in 1:4) {
  arrows(D[i, 1], D[i, 2], D[i+4, 1], D[i+4, 2], col = 4, lwd = 2)
  arrows(D[i, 1], D[i, 2], D[i+8, 1], D[i+8, 2], col = 4, lwd = 2)
}

set.seed(1)
A <- matrix(runif(4 * 2), nrow = 4, ncol = 2)
B <- matrix(runif(4 * 2), nrow = 4, ncol = 2)
C1 <- cbind(B[, 1], A[, 2])
C2 <- cbind(A[, 1], B[, 2])
D <- rbind(A, C1, C2)

plot(A, xlab = expression(x[1]), ylab = expression(x[2]),
     xlim = c(0, 1), ylim = c(0, 1), lwd = 3, cex = 2,
     main = "Sobol' Design (seed 1)", cex.main = 2, cex.lab = 1.5)
points(C1, col = 2, pch = 2, lwd = 3, cex = 2)
points(C2, col = 3, pch = 3, lwd = 3, cex = 2)
for (i in 1:4) {
  arrows(D[i, 1], D[i, 2], D[i+4, 1], D[i+4, 2], col = 4, lwd = 2)
  arrows(D[i, 1], D[i, 2], D[i+8, 1], D[i+8, 2], col = 4, lwd = 2)
}
```






```
par(mfrow = c(1, 1))
```

5 Summary

This chapter covered:

1. **Variance-based sensitivity measures** - First-order and total Sobol' indices
2. **Derivative-based sensitivity measures** - DGSM using finite differences
3. **Morris screening design** - Elementary effects method for factor screening
4. **Sobol' designs** - Designs for variance-based sensitivity analysis
5. **MOFAT designs** - Multi-factor one-at-a-time designs